

The University of Lahore
Department of Computer Science & IT
Mid Term Examination solution



Summer-2023

Course Title:	Programming Fundamentals	Course Code:	CS02115	Credit Hours:	4
Course Instructors:	Dr Nadeem Iqbal, Mr. Waqar, Mr Awais, Mr Majid, Mr Ahemd	Program Name:	BSCS		
Time Allowed:	60 Minutes	Maximum Marks:	25		
Date:	31-10-2023	Course Mentor:	Dr. Nadeem Iqbal		
Student's Name:		Signature:			
Reg. No:		Section:			

Question # 1: Write the Dry run and output or mention error if any. (4*4)=16

Question	Dry Run	Output or Error
<pre>#include <iostream> using namespace std; int main() { long long int _CPU_Register = (10 10) && (20 10); cout<<_CPU_Register; return 0; }</pre>	<pre>(10 10) 1 (20 10) 1 1 & 1 = 1</pre>	1
<pre>#include <iostream> using namespace std; int main() { unsigned long int _ALU_MUX = 1+2/3-1; if(!_ALU_MUX) _ALU_MUX++; for(_ALU_MUX; _ALU_MUX<=1; cout<<_ALU_MUX++, (--+_ALU_MUX)); return 0; }</pre>	<pre>Alu mux = 0 Alu mux = 1 Mux = 2 As its postfix In Memory value of mux is 2 and the output is 1</pre>	1
<pre>#include <iostream> using namespace std; int main(){ for(0;0<=1;1) {cout<<"CPU Dead";} return 0; }</pre>	<pre>Cpu dead Cpu dead Cpu dead Cpu dead . . .</pre>	Infinite lop
<pre>#include <iostream> using namespace std; int main(){ int _RAM_BANK = (true)*(false); for(int i=1;i<=4;++_RAM_BANK,i*=2) for(int k=i;k<=4;++_RAM_BANK,k++); cout<<_RAM_BANK; return 0; }</pre>	<pre>RAM_BANK 0 Outer loop 2 times 3 inc Inner lop 1 4 4 2 4 3 4 4 1</pre>	11

Write a program to draw the following Pattern

```

+/\ /\ /\ /\ /\ /\ /\+
|                               |
+/\ /\ /\ /\ /\ /\ /\+

+/\ /\ /\ /\ /\ /\ /\+
|                               |
|                               |
|                               |
|                               |
|                               |
+/\ /\ /\ /\ /\ /\ /\+

```

Solution 1

```

cout<<"+/\ /\ /\ /\ /\ /\ /\+"<<endl;
cout<<"|                               |"<<endl;
cout<<"+/\ /\ /\ /\ /\ /\ /\+"<<endl<<endl;

cout<<"+/\ /\ /\ /\ /\ /\ /\+"<<endl;
cout<<"|                               |"<<endl;
cout<<"|                               |"<<endl;
cout<<"|                               |"<<endl;
cout<<"|                               |"<<endl;
cout<<"|                               |"<<endl;
cout<<"+/\ /\ /\ /\ /\ /\ /\+"<<endl;

```

Solution 2

```

int width = 12;
int inner_lines=1;
int no_p=2;

for (int pattern = 1; pattern <= no_p; pattern++)
{
    for (int i=1;i<=width;i++)
    {
        if(i==1 || i==width)cout<< "+";
        else if (i%2==0)cout<< "/";
        else cout<< "\\";
    }cout<<endl;
    for(int j=1;j<=inner_lines;j++)
    {
        for (int i=1;i<=width;i++)
        {
            if(i==1 || i==width)cout<< "|";
            else cout<< " ";
        }cout<<endl;
    }
    for (int i=1;i<=width;i++)
    {
        if(i==1 || i==width)cout<< "+";
        else if (i%2==0)cout<< "/";
        else cout<< "\\";

    }cout<<endl<<endl;
    inner_lines+=5;
}

```

Question # 3:**(5 Marks)**

Write a program to get the interger no from the user, and check weather the number is prime or not, if the no is prime then Turn ON the LED, and if the No is not prime then Turn OFF the LED, use SWITCH Statement to handle the both cases.

Example Output:

Enter any interger no : 4

Turning OFF the LED.....

Example Output:

Enter any interger no : 11

Turning ON the LED.....

```
int number ;
cin>>number;
bool is_prime = true;
for ( int i=2 ; i<number; i++)
{
    if (number%i==0)
    {
        is_prime= false;
        break;
    }
}
switch (is_prime)
{
case true:
    cout<<"Turn On the LED.";
    break;
case false:
    cout<<"Turn Off the LED";
    break;
default:
    break;
}
```